What breaks a docking pose

Which ligand properties predict which physical-validity failure, across the PoseBusters benchmark

428 COMPLEXES · 7 DOCKING METHODS · 7,695 POSES

SUMMARY

Docking is scored by whether a predicted pose lands within 2 Å of the crystal structure, but an accurate pose can still be physically impossible. PoseBusters[1] measures this with 18 validity checks and reports failure per *method*. It records nothing about the molecules, so it cannot say which ligands are hard. We computed 27 RDKit descriptors for the 428 benchmark ligands, joined them to all 7,695 published poses, and modelled each check against ligand chemistry.

Failure modes separate by descriptor. Internal-geometry failures (strain energy, self-clash) scale with torsional degrees of freedom; sp^3 chirality failure scales with stereocentre count and with nothing else; cofactor clash scales *inversely* with molecular weight. Protein clash, the largest failure term, admits no ligand-level predictor: its variance is between methods, not between ligands.

Two limits. Only 41 of 126 method × check strata carry enough failures to model, and molecular weight and rotatable-bond count are correlated at $\rho \approx 0.73$–0.74 throughout, so where both are significant they cannot be separated.

1 What breaks which check

Each check was modelled by logistic regression on seven decorrelated descriptors entered together, fitted separately per docking method, with standard errors clustered on ligand. Table 1 lists every descriptor reaching significance for at least two methods. Odds ratios are per standard deviation with the other six held fixed, so a value below 1 means failure becomes *less* likely as the property rises.

TABLE 1 Ligand properties carrying independent signal, per check. Multivariate models, odds ratio per standard deviation, other descriptors held fixed.

CHECK	LIGAND PROPERTY	DIRECTION	METHODS	ODDS RATIO / SD
energy ratio within threshold	n rotatable bonds	↑ raises failure	4 of 5	2.08–7.90
no clashes with organic cofactors	mw	↓ lowers failure	4 of 5	0.17–0.31
no clashes with organic cofactors	n stereocentres	↑ raises failure	4 of 5	2.28–3.15
no internal clashes	n rotatable bonds	↑ raises failure	4 of 5	1.55–9.53
no volume clash with organic cofactors	mw	↓ lowers failure	4 of 5	0.15–0.35
no clashes with organic cofactors	clogp	↑ raises failure	3 of 5	1.54–2.32
no internal clashes	mw	↑ raises failure	3 of 5	2.70–3.35
bond lengths within bounds	mw	↑ raises failure	2 of 2	3.33–47.06
energy ratio within threshold	mw	↑ raises failure	2 of 5	2.14–3.55
no clashes with organic cofactors	n rotatable bonds	↑ raises failure	2 of 5	1.96–2.29
no clashes with protein	clogp	↓ lowers failure	2 of 6	0.56–0.67
no internal clashes	formal charge	↓ lowers failure	2 of 5	0.73–0.82
no internal clashes	n stereocentres	↑ raises failure	2 of 5	1.66–1.79
no volume clash with organic cofactors	n stereocentres	↑ raises failure	2 of 5	3.11–3.55
no volume clash with protein	clogp	↓ lowers failure	2 of 5	0.57–0.65
sp3 stereochemistry preserved	n stereocentres	↑ raises failure	2 of 3	4.33–10.43

Four results follow.

TORSIONAL

Internal-geometry failure scales with torsional degrees of freedom. Rotatable-bond count multiplies failure odds by 1.6–9.5 per standard deviation on both the strain-energy ratio and the internal-clash test, significant in 4 of 5 methods with consistent sign. Under a threefold-minimum torsional model a molecule with k rotatable bonds spans on the order of 3^k conformers; methods that regress coordinates directly, rather than searching torsions of an already-valid conformer, leave that space unconstrained. Both effects hold on the held-out set. OR 1.6–9.5 / SD

STEREOGENIC

sp³ chirality failure scales with stereocentre count, and with no other descriptor. OR 4.3–10.4 per standard deviation; every remaining coefficient exceeds $\alpha = 0.05$ in both fitted methods. If a method assigns each centre independently with accuracy p, then $P(\text{all } n \text{ correct}) = p^n$, so failure probability $1 - p^n$ is monotone increasing in n. The output is a stereoisomer, not a distorted conformer: a distinct chemical entity with distinct activity. OR 4.3–10.4 / SD

INVERSE IN MW	**Cofactor clash is inversely associated with molecular weight.** OR 0.15–0.35 per standard deviation in 4 of 5 methods, so a one-SD decrease in MW multiplies failure odds by roughly 3–7. The distance-based and volume-overlap tests are independent measurements and agree in sign and magnitude per method (TankBind 0.17 and 0.16; EquiBind 0.23 and 0.15). Consistent with steric exclusion: cofactor-occupied volume admits small ligands into contested space, while larger ligands are placed outside it. `OR 0.15–0.35 / SD`
UNPREDICTED	**Protein clash admits no ligand-level predictor.** Fitted on 6 methods (more than any other check) against a base failure rate near 84% for the deep-learning methods. Five of 42 coefficients reach $\alpha = 0.05$ and none replicates beyond two methods. The variance is between methods, not between ligands. `no predictor`

Two descriptors are near-null across the grid: aromatic ring count reaches $\alpha = 0.05$ in 4 of 41 fitted coefficients and halogen count in 2 of 41, against 19 of 41 for molecular weight. Neither predicts any check reproducibly.

Molecular weight is also significant on the internal-geometry checks (3 of 5 methods for self-clash, same sign as rotatable-bond count) but less consistently, and the two are collinear; see §2. Formal charge is inversely associated with self-clash (OR 0.73–0.82, 2 of 5), consistent with intramolecular Coulomb repulsion disfavouring compact conformers.

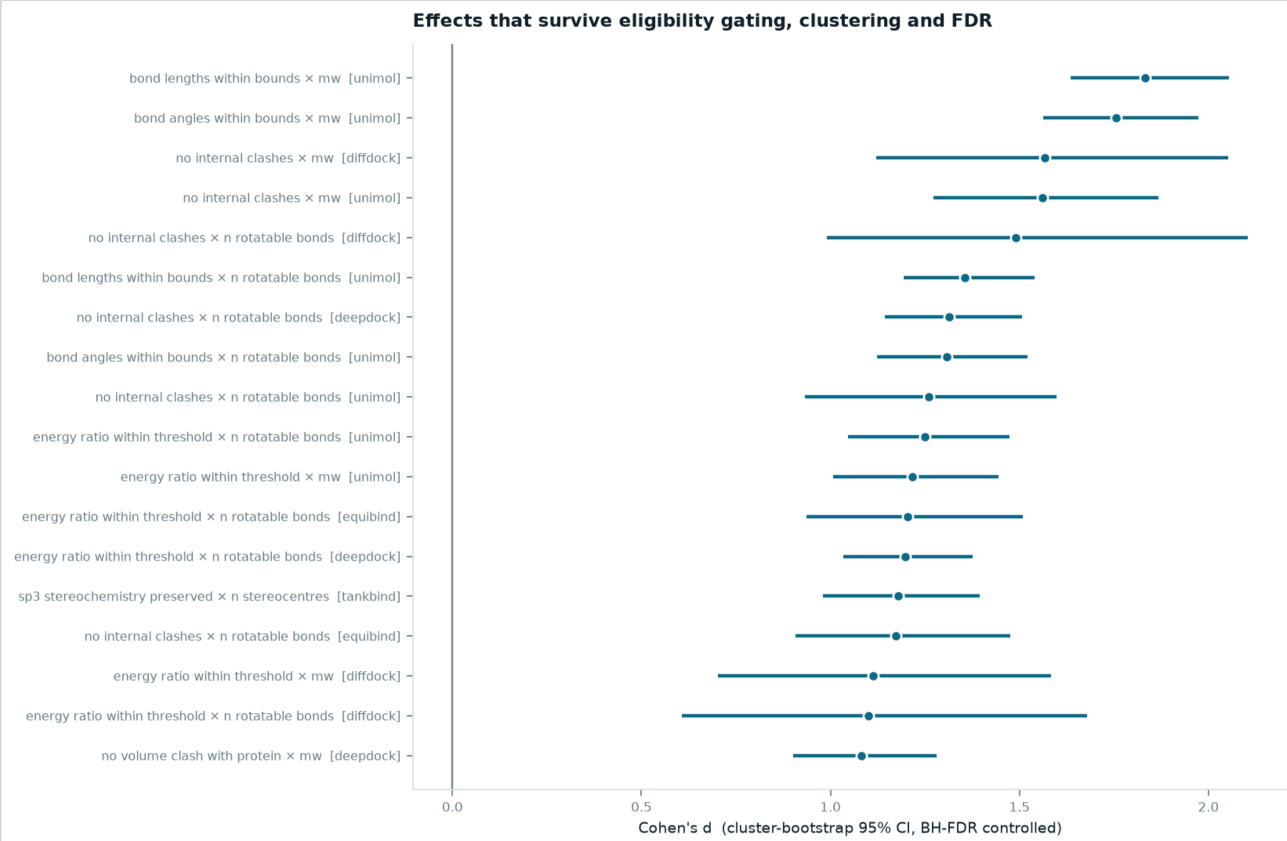

FIGURE 1 The largest single-descriptor associations surviving eligibility gating, ligand-clustered bootstrap intervals and Benjamini–Hochberg correction (95 of 287 estimates survive). Molecular weight and rotatable-bond count recur on the same check for the same method — one signal counted twice, which is why Table 1 uses multivariate models instead.

2 Where size and flexibility cannot be separated

§1 attributes strain and clash failure to torsional freedom rather than bulk. That attribution holds only in the weak form stated. Molecular weight and rotatable-bond count correlate at $\rho \approx 0.73$–0.74 *within every method's ligand set*, not merely on average, so no stratification of these 428 complexes separates them. For DiffDock, the strongest deep-learning method and the usual subject of the question, neither descriptor reaches significance once the other is held fixed ($p = 0.941$ and 0.108), despite both surviving correction individually. Separating them requires a benchmark constructed to decorrelate the two, populating the large-rigid and small-flexible quadrants that this ligand set leaves sparse.

3 Crystal contacts are not the explanation

The benchmark's journal version drops 120 of 428 complexes whose ligand touches a crystallographic symmetry mate, on the reasoning that such contacts inflate protein-clash failure. That subset was never published, so we recomputed it from spacegroup expansion: 104 of 427 resolvable complexes have a protein contact within 4.0 Å (119 counting solvent), bracketing the authors' 120. Contact status does not account for the clash rate. One method of seven shows a significant difference and it runs in the opposite direction: Uni-Mol's clash failure is 17.1 points lower among contact-bearing complexes (95% CI −27.0 to −7.7). The other six straddle zero; Vina and Gold fail so rarely (2 of 323 and 16 of 322 poses) that their point estimates carry no information.

4 How large the gap is

Classical search methods are accurate on about half of complexes and valid on 94–97% of all poses. The deep-learning methods are less accurate and, conditional on accuracy, predominantly invalid.

TABLE 2 Accuracy, validity and their intersection. Raw poses, 428 complexes.

METHOD	CLASS	RMSD ≤ 2 Å	VALID	BOTH	OF ACCURATE POSES, INVALID
vina	classical	52.3%	97.4%	51.2%	2.2%
gold	classical	51.2%	93.9%	47.7%	6.8%
diffdock	deep learning	37.9%	24.1%	13.6%	64.2%
unimol	deep learning	22.9%	5.6%	1.9%	91.8%
deepdock	deep learning	17.8%	8.2%	4.7%	73.7%
tankbind	deep learning	15.0%	4.7%	2.6%	82.8%
equibind	deep learning	2.6%	0.9%	0.2%	90.9%

Reporting invalidity only among accurate poses conditions on an outcome that shares an upstream cause with validity. Banding raw RMSD instead uses every pose and needs no such conditioning — and shows the failure is not confined to poses near the threshold. Deep-learning validity is already poor at sub-Ångström accuracy (DiffDock 61.3%, Uni-Mol 33.3%) and does not recover further out; the classical methods stay above 88% in every band.

TABLE 3 Share of poses passing all applicable checks, by RMSD band. Unconditioned on accuracy.

METHOD	(0.0, 1.0]	(1.0, 2.0]	(2.0, 3.0]	(3.0, 5.0]	(5.0, 10.0]	(10.0, INF]
deepdock	55.0%	16.1%	12.5%	6.8%	1.1%	0.0%
diffdock	61.3%	20.0%	20.0%	17.8%	18.6%	12.7%
gold	95.2%	88.9%	95.0%	90.7%	97.6%	100.0%
tankbind	36.4%	13.2%	3.0%	1.8%	0.0%	6.9%
unimol	33.3%	2.5%	6.2%	5.2%	4.2%	0.0%
vina	100.0%	93.5%	97.6%	96.2%	96.2%	100.0%

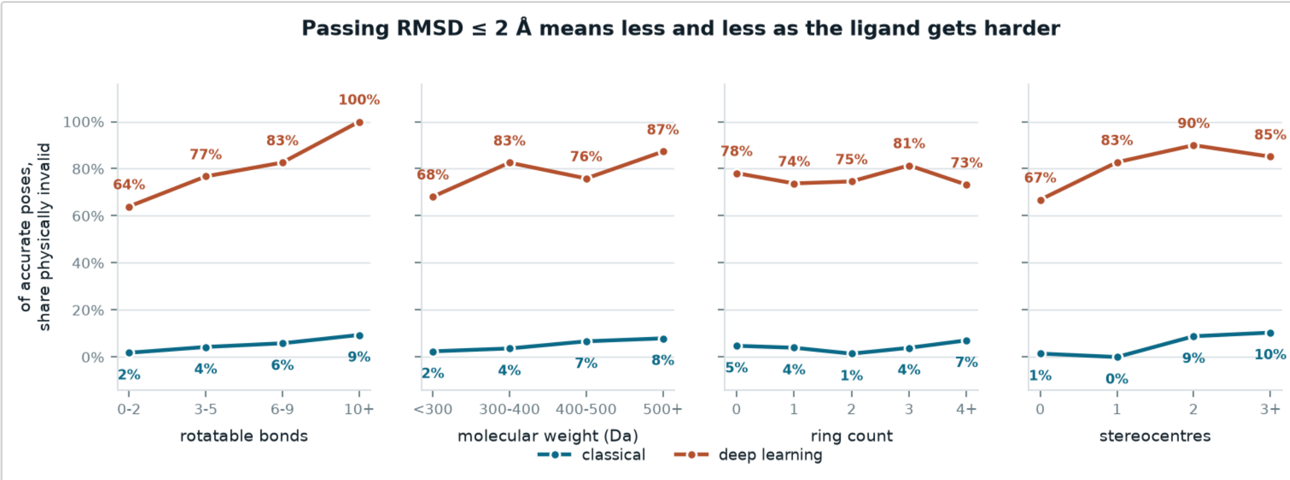

FIGURE 2 Share of RMSD-accurate poses that are physically invalid, across four ligand partitions. Monotone against rotatable-bond count in both method classes, close to flat against ring count.

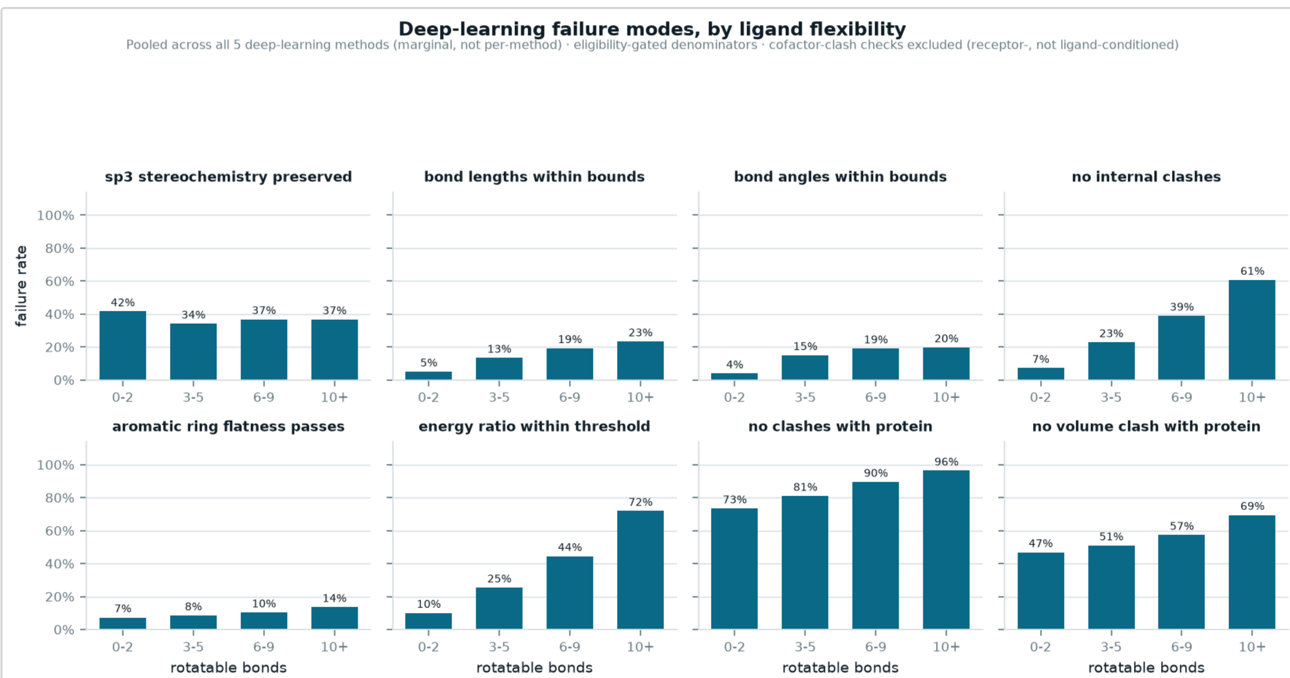

FIGURE 3 Per-check failure against rotatable-bond count, pooled across the five deep-learning methods, with eligibility-gated denominators. Strain and internal clash climb steeply; planarity checks barely move. Cofactor checks are excluded as receptor-conditioned.

5 Confidence in these numbers

The published checks reproduce. Everything here reads the paper's published check outcomes rather than recomputing them, so we re-ran PoseBusters on all 513 crystal ligands against their own receptors: worst-case agreement `99.81%` across 17 checks, with 15 agreeing on all 513 structures. The two disagreements are single structures on different checks.

The main effects replicate held-out. The 85-complex Astex Diverse Set was untouched throughout. Of 14 shortlisted effects, 8 replicate with both intervals excluding zero — the strain and internal-clash effects against rotatable bonds hold for every deep-learning method tested; the chirality and protein-clash effects do not.

Two thirds of the grid is unmeasurable. 85 of 126 method × check strata could not be estimated, 59 of them because the check never failed for that method. These are absent measurements rather than null results; an effect grid reported without that denominator conflates the two.

6 What this means in practice

> Three descriptors carry the predictive signal, each specific to a failure mode: rotatable-bond count for strain and self-clash, stereocentre count for chirality, molecular weight for cofactor clash in the inverse direction. No single descriptor predicts validity in aggregate, so a scalar difficulty score over these ligands would average out check-specific effects of opposite sign.
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> The dominant failure term is not ligand-dependent. Protein clash accounts for the largest share of deep-learning invalidity and is unpredicted by any descriptor measured here, so ligand-level triage cannot reduce it; that variance is attributable to the method.

7 Method

27 RDKit[4] descriptors per crystal ligand (size, flexibility, ring system, stereochemistry, polarity, composition), joined to the published results table[2] on PDB code. Blank check values in that table mean the check was *not run*, not that it failed. Both flatness columns are blank for all 2,996 energy-minimised rows, and 85 rows have every check blank because the method produced no pose; treating either as a pass inflates validity.

Each check is restricted to ligands that could fail it: a molecule with no stereocentres cannot fail the chirality check, and counting it as a trivial pass both deflates the failure rate and inflates any effect measured against the gating descriptor. Gating raises the observed chirality-failure rate from 21.4% to 36.7%. The four cofactor checks are further restricted to cofactor-bearing complexes, since they pass almost automatically otherwise (2.4% versus 22.5% failure).

Effects are estimated per method, never pooled. Intervals come from a bootstrap resampling *ligands* rather than poses, since each ligand contributes one pose per method and treating 2,140 rows as independent would report intervals roughly √5 too narrow. Strata with fewer than 15 failures or 30 distinct ligands are not estimated. Single-descriptor estimates across the whole grid are corrected by Benjamini–Hochberg[5] at $\alpha = 0.05$ as one family; because they cannot separate correlated descriptors, the per-check conclusions in §1 come from the multivariate models. Crystal contacts were computed with gemmi[6] by expanding each deposited structure[7] by its spacegroup.

Caveats. Descriptors describe the docked molecule, not the returned pose. The predicted poses were never released, so only the crystal-structure rows could be independently recomputed. Several surviving effects rest on small failure counts even after gating (DiffDock's internal-clash effects rest on 25), so single rows should not be read as decisive.

References

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2. Buttenschoen M, Morris GM, Deane CM. PoseBusters paper data. Zenodo (2023). doi:10.5281/zenodo.8278563
3. RDKit: open-source cheminformatics. rdkit.org
4. Benjamini Y, Hochberg Y. Controlling the false discovery rate. *J R Stat Soc B* **57**, 289–300 (1995).
5. Wojdyr M. GEMMI: a library for structural biology. *J Open Source Softw* **7**, 4200 (2022).
6. Berman HM *et al.* The Protein Data Bank. *Nucleic Acids Res* **28**, 235–242 (2000).

Every table is read from the analysis outputs at render time and every figure inlined from its original, so this document cannot diverge from the analysis it reports. Deterministic under fixed seeds; 37 tests. Regenerate with `python -m pb.paper_html`.